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Chemistry Paper 3B WASSCE (SC), 2023

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Question 2B

G is an inorganic salt. Carry out the following exercises on G

Record your observations and identify any gas(es) evolved.

State the conclusions you draw from the results of each test.

(a) Dissolve all of G in about 10 cm³ of distilled water in a boiling tube

(b) (i) Test the resulting solution with litmus paper.

(ii) To about 2 cm³ portion of the solution in a test tube, add NaOH(aq). Then warm the resulting mixture gently.

(iii) To another 2 cm³ portion of the solution, add dilute HNO₃.

(iv) To 2cm³ portion of the solution, add BaCl₂(aq) followed by excess dilute HNO₃.

Observation

The performance of the candidates was below average in this question.

In part (a), majority of the candidates dissolved G in about 10 cm³ of distilled water in a boiling tube.

In part (b), majority of the candidates did not adhere to the instructions, and this affected their performance.

The expected answers include:

	TEST	OBSERVATION	INFERENCE
(a)	G + distilled water	Colourless solution obtained	
(b)	Solution of G + litmus papers	Turn red litmus blue	G(aq) is alkaline/basic

(c)	Solution of G + NaOH(aq) Mixture warmed	No precipitate formed / no visible reaction Colourless gas with pungent /chocking / irritating smell evolved Gas turned red litmus paper to blue / formed dense white fumes with conc HCl	Gas is NH ₃ from NH ₄ ⁺
(i)	Solution of G + HNO ₃ (aq)	Effervescence, gas evolved colourless, odourless Gas turned lime water milky	Gas is CO ₂ from CO ₃ ²⁻ / HCO ₃ ⁻ present
(ii)	Solution of G + BaCl ₂ (aq) + HNO ₃ (aq)	White (Chalky) precipitate formed Precipitate dissolved (with rapid effervescence)	CO ₃ ²⁻ - SO ₃ ²⁻ - SO ₄ ²⁻ - CO ₃ ²⁻ present

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